SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

GreenLoop

- TEAM NAME : Pentaforce
- THEME: Sustainability Development
- PS CATEGORY- Software
- TEAM ID-DEX26266

2026

IDEA : GREENLOOP

THE PROBLEM : Campuses and housing communities generate emissions from power, waste and water, but nobody can see them in one place. Without visibility there is no way to find the biggest savings, and no motivation to change.

1

TRACK

Energy, solar, waste and water data in one dashboard.

2

ANALYZE

Everything is converted into a single unit, kg CO_2e.

3

RECOMMEND

AI ranks the top actions by kg CO_2 saved per week.

4

REWARD

Block-vs-block leaderboard and green points keep people engaged.

How it addresses the problem

- Live kg CO_2e per block, hostel or home
- Pinpoints the biggest savings, not generic tips
- Rewards action, so behaviour changes

What makes it unique

- Automatic and continuous, not a one-time calculator
- Community-level, not just individual

Starts on simulated data; hardware optional

TECHNICAL APPROACH

1. Data inputs
ESP32 sensor or simulated data; waste photo

2. ML classifier
MobileNetV2 on TrashNet / TACO

3. Emission engine
kWh and kg waste to kg CO_2e

4. Recommender
Ranks top 3 actions by CO_2 saved

5. Dashboard + rewards
Live score, leaderboard, points

TECHNOLOGIES USED

Frontend	React + Tailwind (or Streamlit)
Backend+ DB	FastAPI / Flask + SQLite or Supabase
ML	MobileNetV2 on TrashNet / TACO, 5 waste classes
IOT	ESP32 + PZEM-004T over MQTT (simulated in demo)
Charts	Recharts / Chart.js / Plotly

CORE CALUCULATIONS

Electricity: kWh × grid factor (CEA)
Solar offset: solar kWh × grid factor
Waste: kg × factor per category
Net score: emissions − avoided
Saving: (current − optimized) × factor
Every factor is cited in-app.

FEASIBILITY AND VIABILITY

Why it is feasible

- Software-first, runs on simulated data
- Pretrained MobileNetV2, public emission factors
- Optional low-cost ESP32 sensors
- Free-tier cloud, open-source stack
- Pilot-ready: one campus first

CHALLENGES	HOW WE OVERCOME IT
Classifier misreads on stage	Test on-stage items,report real accuracy, keep a backup video
Hardware fails or time runs out	Simulated data first, ESP32 Last
Credibility of CO2 numbers	Cited Factors (CEA ,IPCC, EPA, WARM) in-app
Will people change behaviour?	Quatified tips, Leaderboard, Reward
Scope creep	Must have first; skip accounts , payments

BUSINESS CANVAS

Mission and vision

• Mission: make every footprint visible and every saving actionable
• Vision: a measurable path to net zero for every campus and village

Market and competitive analysis

•Customers: colleges, apartments, panchayats, factories
•Existing calculators are individual, manual, one-time
•GreenLoop is automatic, continuous, community-level

Business perspective

•Revenue: subscription for institutions
•Paid hook: ESG reporting
•Go-to-market: one campus pilot, then residential

Financial planning and requirements

•MVP on free-tier cloud, open-source stack
•Optional ESP32 sensor node, low cost per site
•Per-site pricing, validated in the pilot

IMPACT AND BENEFITS

WHO BENEFITS Students, wardens, residents, RWAs, panchayats, administrators

SOCIAL

- Builds climate awareness through live feedback
- Peer competition builds green habits
- Works for campuses, apartments, villages

ECONOMIC

- Lower power and water bills
- Less waste-handling cost
- ESG-ready reports

ENVIRONMENTAL

- Cuts CO_2e from power, waste, pumping
- More solar, more avoided emissions
- Less waste to landfill

SDG 7
Clean energy

SDG 11
Sustainable communities

SDG 12
Responsible use

SDG 13
Climate action

SDG 6
Clean water (secondary)

RESEARCH AND REFERENCES

1. CEA CO_2 Baseline Database, Indian Power Sector
   cea.nic.in
2. 2006 IPCC Guidelines, Vol. 5: Waste
   ipcc-nggip.iges.or.jp/public/2006gl/vol5.html
3. US EPA Waste Reduction Model (WARM)
   epa.gov/warm
4. TrashNet waste image dataset
   github.com/garythung/trashnet
5. TACO: Trash Annotations in Context
   tacodataset.org
6. MobileNetV2 (Sandler et al., 2018)
   arxiv.org/abs/1801.04381
7. UN Sustainable Development Goals
   sdgs.un.org/goals

GITHUB REPOSITORY

OUR COMMITMENTS

- Demo uses simulated energy data
- Figures are stated estimates
- Accuracy from our own test set